MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

FULL STACK DEVELOPER

MASTER PROGRAM

Build End-to-End Web Applications from Frontend to Backend

COURSE HIGHLIGHTS

- HTML, CSS & JavaScript
- React.js / Angular
- Python / Java & MySQL
- REST API & Deployment

FULL STACK WEB DEVELOPMENT

Build • Code • Deploy • Grow



COURSE OVERVIEW

A practical, career-focused program covering frontend development, modern frameworks, backend programming, databases, REST APIs, and complete application integration.

KEY TOPICS

1 Frontend Development

- HTML5 & CSS3
- Responsive web design
- JavaScript & DOM
- React.js / Angular
- API integration

2 Backend Development

- Python / Java
- OOP concepts
- Flask / Java web development
- REST API development
- Authentication & sessions

3 Database

- MySQL & SQL
- CRUD operations
- Joins & relationships
- Database design

4 Full-Stack Integration

- Frontend–backend communication
- JSON & REST APIs
- Git & GitHub
- Testing & deployment

WHO CAN JOIN?

- Students & graduates
- Beginners in programming
- Aspiring web developers
- Career changers
- Working professionals
- Freelancers & startup enthusiasts

LEARNING OUTCOMES

By the end of the course, learners can:

- Build responsive web applications
- Develop frontend applications using React.js / Angular
- Create backend applications using Python / Java
- Develop and integrate REST APIs
- Design and manage MySQL databases
- Perform CRUD operations
- Connect frontend, backend & database
- Use Git & GitHub
- Test, debug and deploy applications
- Build a complete full-stack project

SKILLS TODAY
FOR A SMARTER TOMORROW

FRONTEND DEVELOPMENT

Create responsive, interactive and user-friendly web interfaces



- BUILD RESPONSIVE WEBSITES
- MODERN UI/UX DESIGN PRINCIPLES
- INTERACTIVE WEB APPLICATIONS
- INDUSTRY-RELEVANT SKILLS
- BUILD A STRONG FRONTEND DEVELOPER FOUNDATION

TURN IDEAS INTO BEAUTIFUL WEBSITES

FRONTEND DEVELOPMENT

HTML

Web Structure & Content

- HTML structure and syntax
- Tags, elements & attributes
- Forms & input types
- Tables & lists
- Semantic HTML5
- Audio, video & iframe
- SEO-friendly HTML structure

CSS

Styling & Responsive Design

- CSS syntax & selectors
- Combinators & specificity
- Box model
- Margin, padding & borders
- Flexbox
- CSS Grid
- Responsive design
- Media queries
- Transitions & transforms
- CSS animations
- Bootstrap basics

JAVASCRIPT

Interactive Web Development

- Variables & data types
- Operators
- Conditions & loops
- Functions & scope
- Arrays & objects
- DOM manipulation
- Event handling
- ES6+ features
- Form validation
- LocalStorage
- Fetch API & AJAX

KEY SKILLS

Structure → Style → Interactivity → Responsive Design

FRONTEND FRAMEWORK & BACKEND DEVELOPMENT

FRONTEND FRAMEWORK

Choose One: **React.js** OR **Angular**

REACT.JS

- Component-based architecture
- JSX & React elements
- Props & State
- Event handling & forms
- React Hooks
  useState • useEffect • useContext
- React Routter
- API integration
- Conditional rendering
- Lists
- Project structure & deployment

ANGULAR

- Angular architecture
- Angular CLI
- Components, modules & services
- One-way & two-way binding
- Directives & pipes
- Routing & navigation
- HTTP Client
- REST API integration
- Reactive & template-driven forms
- Dependency Injection
- Application deployment

BACKEND DEVELOPMENT

Choose One: **Python** OR **Java**

PYTHON

- Python syntax & data types
- Conditions & loops
- Functions, modules & packages
- OOP concepts
- Classes & inheritance
- Exception handling
- File operations
- Flask web development
- Routing, templates & sessions
- Forms
- REST API development
- MySQL connectivity

JAVA

- Java fundamentals
- OOP principles
- Classes & methods
- Exception handling
- Collections Framework
- JDBC
- Servlets & JSP
- Form handling
- Session management
- MySQL integration

BUILD MODERN WEB APPLICATIONS

DEVELOP SCALABLE SOLUTIONS

GAIN PRACTICAL SKILLS

BECOME INDUSTRY-READY

SKILLS TODAY
FOR A BRIGHTER TOMORROW

DATABASE & API INTEGRATION

1 DATABASE — MYSQL

MANAGE AND ORGANIZE APPLICATION DATA
- Data storage
- Data retrieval
- Data security
- Backup & recovery

SQL & DATABASE FUNDAMENTALS
- RDBMS concepts
- Database & table creation
- SQL fundamentals
- SELECT
- INSERT
- UPDATE
- DELETE

ADVANCED SQL
- Joins
- Subqueries
- Views
- Indexes
- Data constraints
- Normalization
- Stored procedures — Basics
- Triggers — Basics

DATABASE DESIGN
- Schema design
- Primary & foreign keys
- Relationships
- Data integrity
- CRUD operations

2 REST API & APPLICATION INTEGRATION

API DEVELOPMENT
- REST architecture
- HTTP methods
- Request & response
- JSON data
- CRUD APIs
- API integration

FULL-STACK CONNECTION



3 KEY CONCEPTS





PRACTICAL LEARNING

REAL-TIME SCENARIOS

HANDS-ON CONFIGURATION

EXPERT GUIDANCE

BUILD YOUR DATABASE & API SKILLS FOR A CONNECTED FUTURE

DATA SCIENCE WITH GEN AI

PRACTICAL SKILLS + REAL PROJECTS + FUTURE READY

PROJECT, TOOLS & DEPLOYMENT

PROJECT & FINAL INTEGRATION

- Build a Complete Full-Stack Application
- CAPSTONE PROJECT
- DEVELOPMENT PROCESS

Plan → Design → Develop → Integrate → Test → Deploy

PROJECT COMPONENTS

Frontend
- Responsive UI
- Components
- Forms
- User interaction

Backend
- REST APIs
- Business logic
- Authentication
- Session handling

Database
- MySQL schema
- CRUD operations
- Data relationships

Integration
- API communication
- JSON data transfer
- Frontend-backend connection

TESTING & VERSION CONTROL

APPLICATION TESTING

- Functional testing
- API testing
- Error handling
- Debugging
- Data validation

GIT & GITHUB

- Git fundamentals
- Repository management
- Commit & push
- Branching
- Version control
- Project collaboration

DEPLOYMENT

DEPLOY YOUR APPLICATION

- Frontend deployment
- Backend deployment
- Production basics

- Application hosting
- Netlify
- Render / Heroku

TECHNOLOGY STACK

HTML | CSS | JavaScript | React.js / Angular | Python / Java | MySQL | REST API | Git & GitHub

FINAL OUTCOME

Build → Integrate → Test → Deploy → Showcase

>>> YOUR FULL-STACK DEVELOPMENT JOURNEY STARTS HERE. >>>



- PRACTICAL LEARNING
- REAL-TIME SCENARIOS
- HANDS-ON PROJECTS
- CAREER GROWTH

SKILLS TODAY FOR A SMARTER TOMORROW